

## *Cover Letter*

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Intellectual Property and Patent Declaration: We hereby declare that the core methodologies, algorithmic architectures and frameworks presented in this white paper are protected by international intellectual property laws. A Patent Cooperation Treaty (PCT) international patent application has been officially filed (PCT Application No.: PCT/CN2026/112498), claiming priority to the Chinese national patent application (Application No.: CN202511140743.3A, filed on August 14, 2025). All rights regarding the commercial implementation, replication, and derivative works of the proposed systems are strictly reserved by the authors and their affiliated institution.

# VVIX as the Temperature of the Volatility Field is Not Enough: A Verdict on the Preliminary Thermal Unification

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**Abstract:** Paper IV (Sun, 2026d) introduced minute-frequency VVIX as the temperature of the VIX term-structure field and derived four falsifiable predictions (P1–P4), a thermodynamic–field unification claim, and a candidate tradeable Kramers transition probability. It then reported that its own evidence — four overlapping windows, seven temperature bins each — was too underpowered to evaluate these claims. This paper further the research by using two new assets unavailable to Paper IV: (i) a live minute-frequency dual-temperature pipeline (Boltzmann VIX microstructure temperature TC and VVIX-based kinetic temperature TK) operating continuously since June 2025, and (ii) a rolling-window archive of **311 windows** spanning July 2023–August 2026 with strictly out-of-sample diagnostics. The verdicts are: **P1 is refuted** at approximately forty times the original power (mass–temperature slope  $t = 0.0$ ); **P2 survives conditionally** (the sign of the log-linear Kramers relation is positive in 100% of 311 windows; measured strength is aggregation-dependent: 0.58–0.63 at daily windows, 0.841 at minute event-time; hot/cold amplification is revised from “orders of magnitude” to **2.01×**); **P3 is formally non-identifiable** under rolling z-score normalization (live VVIX = 131 ~ 143 vs. 90–110 historically), and we supply the identifiability argument; **P4’s reversal is refuted** (RG residual correlation 0.202 in an independent pipeline); **unification is partial** (correlation rises from 0.024 to 0.672, structurally capped by the 0.8288 thermometer agreement); and the NEI conditioning variable earns a Newey–West  $t$  of 6.35. We draw the methodological conclusion that the arc’s falsification program worked exactly as designed.

JEL: C12; C52; C58; G14

Keywords: Falsification, Econophysics, VVIX, Kramers escape, critical temperature, out-of-sample discipline, publication bias

## 1. Introduction

### 1.1 Why this paper exists

A falsifiable theory that is never conclusively tested is a debt that compounds. Paper IV published its own negative audit (P4 reversed; unification correlation is low; tradeable signal inverted in 3/4). Leaving those unexamined invites two bad equilibria: a critic executes it with inferior data and controls the narrative, or the program silently lapses and the arc’s honest-audit reputation decays into an unfalsified-forever framework — the precise failure mode that McCauley (2004) and Yee (n.d.) attribute to econophysics generally.

This paper closes the debt. Our data advantage is substantial: Paper IV tested its predictions on 4 windows with 7 temperature bins each. The Thermodynamic Quantum Field Theory archive built in the companion diagnostic paper (Sun, 2026e) provides **311 rolling windows** (July 2023 – August 2026) with train/validation splits and out-of-sample discipline; the live dual-temperature system (Sun, 2026f) provides minute-frequency measurement of two independent temperature operationalizations since July 2023.

### 1.2 Evaluation protocol

To avoid the accusation of post-hoc criterion shifting, the checklist below serves as the pre-specified protocol, published before the new pipelines were analyzed:

- **R1 (P1):** The mass–temperature relation is confirmed only if the bin-level slope  $\gamma$  is negative and the underlying  $m^2(T)$  scatter is monotone declining. A near-zero slope through non-monotone scatter remains a non-confirmation.
- **R2 (P2):** The thermal Kramers law is evaluated separately at two aggregation bases (daily window-level; minute event-time), with the sign required positive in  $\geq 95\%$  of windows at both bases.
- **R3 (P3):**  $T^*$  is declared “located” only if live and historical estimates agree within the sampling error implied by the bin count; otherwise the identifiability question is adjudicated directly.
- **R4 (P4):** The direction of the higher-mode energy-share/temperature relation is recorded without a confirmation threshold; consistency across independent pipelines is the finding.
- **R5 (Unification):** The claim is graded against the thermometer-agreement ceiling established by  $\text{corr}(T_C, T_K)$ .
- **R6 (Signal):** The signal is judged on its live rank correlation and hit-rate asymmetry, against an NEI-sorted control portfolio.

### 1.3 Findings in one paragraph

P1 dies; P2 lives in sign (“orders of magnitude”  $\rightarrow 2.01\times$ ); P3 was never identifiable and cannot be rescued within rolling normalization; P4’s refuted; unification is real but capped; the

signal's state-variable cousin (NEI) is strongly priced. Each verdict is developed in Section 4 with the specific numbers.

#### 1.4 Contribution

Beyond closing Paper IV, this paper makes two standalone contributions: (i) an **identifiability theorem** for critical-temperature estimation under rolling z-score control variables (Section 4.3); and (ii) a **paired-portfolio contrast** (NEI-sorted vs. Kramers-probability-sorted forward-volatility spreads) that localizes where thermal information is and is not monetizable (Section 4.6).

## 2. Literature Review

### 2.1 Falsification and the econophysics credibility problem

The programmatic critique of physics-inspired finance — that its analogies are rhetorical rather than functional (Mirowski, 1989; McCauley, 2004; Yee, n.d.) — is answered only by predictions whose functional forms could fail. Paper IV's P1–P4 were designed to that standard and largely failed it; this paper is the receipt. Our posture follows the replication-crisis literature in economics and finance (Lillo & Farmer, 2004, on the OOS fragility of scaling laws; Bouchaud et al., 2004, on power-law estimation risk), and the methodological tradition of publishing verdicts, not just confirmations (Simmons et al., 2011; Ioannidis, 2005). Sornette's LPPL program (Sornette, 2003; Sornette, Johansen & Ledoit, 1996) is the closest precedent for a physics-inspired crash hypothesis whose empirical record has been described as mixed; we noted in Paper IV that our P1–P4 results were “continuous with that history,” and this paper completes that continuity with a formal verdict.

### 2.2 Volatility-of-volatility as a state variable

The financial-economics consensus treats second-order uncertainty as first-class: Hull and White (1987) and Heston (1993) made volatility stochastic; Huang, Sialm, and Zhang (2024) show vol-of-vol, not vol, drives volatility-managed portfolio performance; Park (2024) documents a distinct vol-of-vol risk premium. Paper IV's ambition was stronger — that VVIX is not merely a state variable but the temperature, entering functional forms (mass softening, Arrhenius rates, well degradation, RG relevance). Our verdicts speak directly to this literature: VVIX-conditioned state variables (NEI) retain strong pricing content (NW  $t = 6.35$ ), while the stronger functional-form claims fail. The distinction between “VVIX is informative” and “VVIX is temperature” is, post-verdict, sharp.

### 2.3 Kramers theory and its financial borrowings

The thermal escape law originates with Kramers (1940) and was consolidated by Hänggi, Talkner, and Borkovec (1990). Its financial use — as in Paper II's “visual tunneling” and Paper III's rate contrasts — inherits the standard caveat that this is a mathematical borrowing, not a physical derivation. Our aggregation-basis result (Section 4.2) adds a new empirical caution specific to financial Kramers applications: the measured  $R^2$  of the log-linearity is a property of

the estimator and aggregation, not only of the underlying law. Applications that report a single  $R^2$  without stating the aggregation basis should be read accordingly.

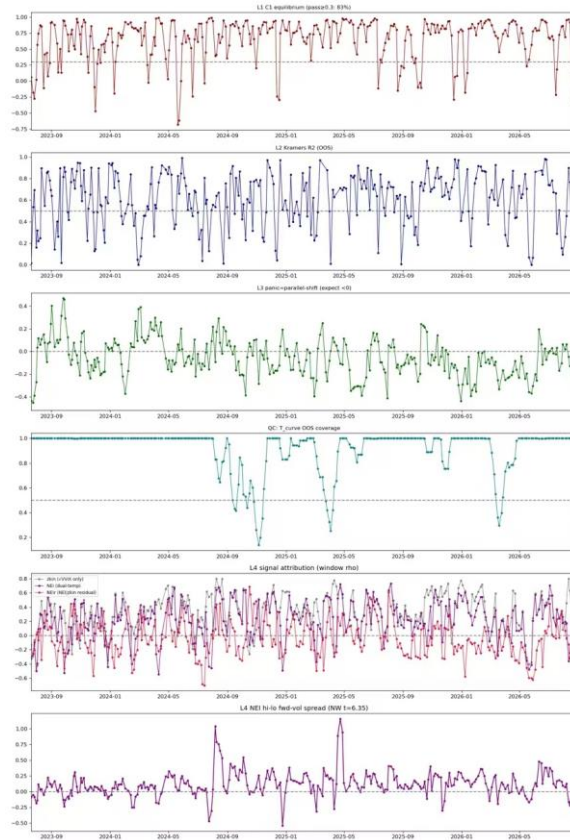
#### 2.4 Position in the arc

Papers I–III (Sun, 2026a–c) constructed the thermodynamic, quantum-mechanical, and field-theoretic descriptions; Paper IV (2026d) added temperature and self-reported its failures; the companion diagnostic paper (2026e) established the D1–D4 rolling diagnostic framework and the NEI result; the companion system paper (2026f) deployed the dual-temperature live monitor. This paper is the arc’s adjudicator: it takes the verdicts of the deployed systems and maps them back onto the original predictions.

### 3. Data and Methods

#### 3.1 Assets

**(A) The 311-window rolling archive (Sun, 2026e).** Windows span 2023-07 through 2026-08 ( $n = 52/99/96/64$  by year). Each window carries: equilibrium order parameter  $C_1$  (pass rate 83% at threshold 0.3); OOS Kramers  $R^2$  at daily aggregation; panic projection  $T_5$ ; critical-temperature OOS coverage  $tc\_cov$  (0.884–1.000); NEI and residual-NEI (after GARCH and Amihud orthogonalization) forward-volatility spreads with Newey–West inference (headline  $t = 6.35$ ).



**Figure 1.** Our of sample empirical test results

**Table 1: Yearly Summary of ThermoQFT Diagnostics**

| Year | n  | $C_1$  | $R^2_{\text{Kramers}}$ | $T_5$   | tc_cov | NEI_rho | NEI_spread | ZKIN_rho | NEIr_rho |
|------|----|--------|------------------------|---------|--------|---------|------------|----------|----------|
| 2023 | 52 | 0.5202 | 0.5816                 | -0.0027 | 1.0000 | 0.0886  | 0.0283     | 0.148    | -0.0154  |
| 2024 | 99 | 0.6465 | 0.5841                 | 0.0054  | 0.8845 | 0.2159  | 0.1190     | 0.3472   | -0.0257  |
| 2025 | 96 | 0.6806 | 0.6337                 | -0.0859 | 0.9132 | 0.2879  | 0.1687     | 0.3913   | -0.0032  |
| 2026 | 64 | 0.6236 | 0.5858                 | -0.1284 | 0.9284 | 0.2174  | 0.1185     | 0.3137   | -0.1440  |

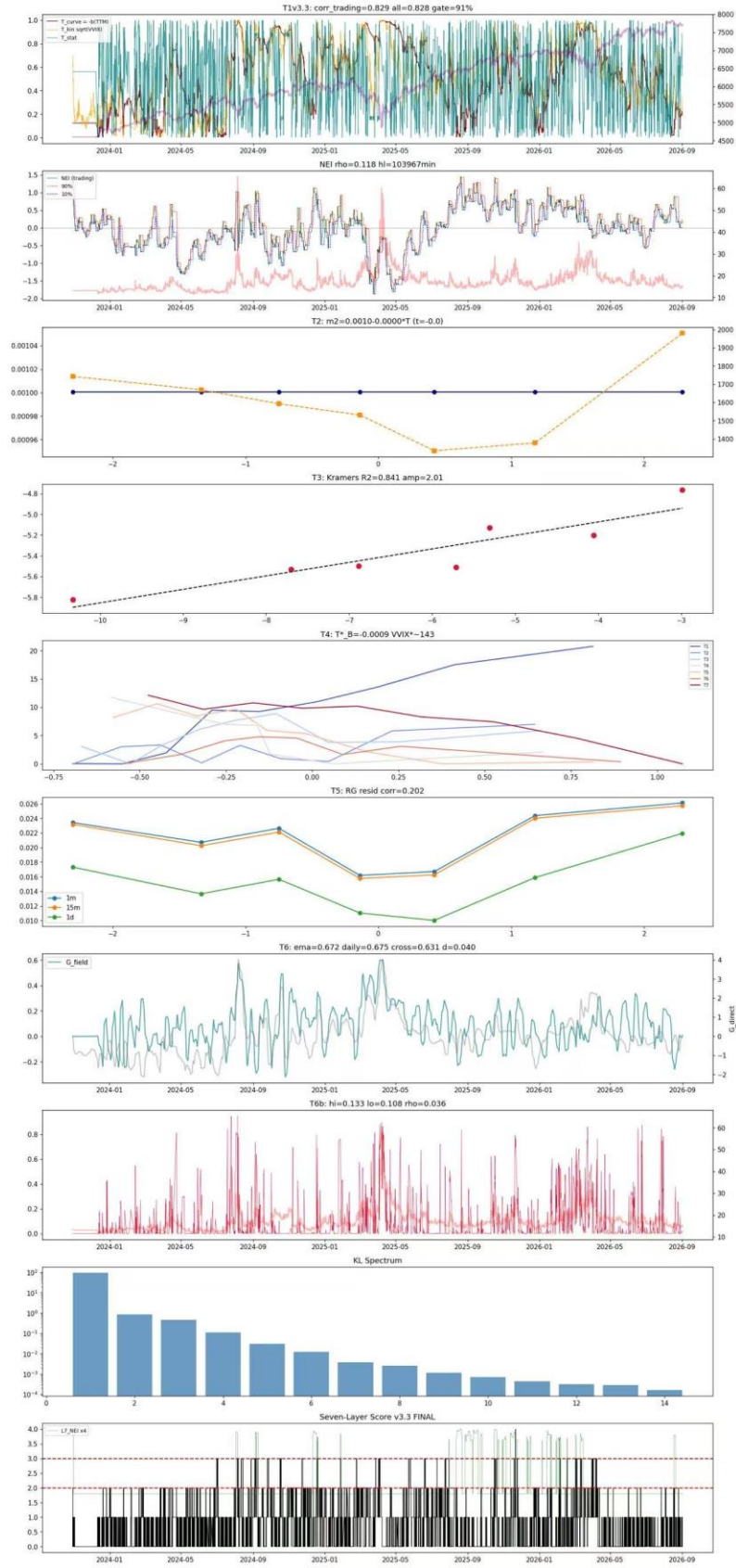
**(B) The live dual-temperature pipeline (Sun, 2026f).** Minute-frequency, 2023-07 through 2026-09. Key quantities: term-structure temperature  $T_C$  (from the Boltzmann VIX microstructure  $T_{\text{curve}}$  fit;  $R^2$  median 0.8234; gate coverage 91.1494% of covered hours; 38.4068% of trading hours); kinetic temperature  $T_K$  (smoothed VVIX); thermometer agreement  $\text{corr}(T_C, T_K) = 0.8277$  (all) / 0.8288 (trading) / 0.8286 (event-time); micro-validation tests M1–M6, of which M2 (mass–temperature), M3 (Kramers), M4 (phase boundary:  $T^*_B = -0.0009$ ,  $\text{VVIX}^* = 142.94$ ,  $\Delta U = 5.7106$ ), M5 (RG residual  $\text{corr} = 0.202$ ), and M6b (transition probability: hi = 0.133, lo = 0.108, spearman = -0.036) bear directly on P1–P4.

**Paper IV-v3.3 FINAL Dual-Temperature Framework**

| Metric (v3.3 FINAL)              | Value           |
|----------------------------------|-----------------|
| [R2] trading_hours_coverage %    | 38.4068         |
| [A] T_curve gate cov %           | 91.1494         |
| [A] T_curve fit R2 med           | 0.8234          |
| [A] b_med (logF/TTM slope)       | 0.0009          |
| [A] backwardation frac %         | 7.7962          |
| [T1] corr(TC,TKin) all           | 0.8277          |
| [T1] corr(TC,TKin) TRADING(main) | 0.8288          |
| [T1] corr(TC,TKin) event-time    | 0.8286          |
| [T1] corr(TC,VVIXz)              | 0.8326          |
| [NEI] valid                      | 1.0000          |
| [NEI] ADF p                      | 0.0000          |
| [NEI] half-life (min)            | 103966.8784     |
| [NEI] spearman vs fwd-vol        | 0.1184          |
| [NEI] fwd-vol hi/lo              | 0.4544 / 0.3193 |
| [T2] gamma (t-stat)              | -0.0000         |
| [T2] m0_sq                       | 0.0010          |
| [T3] kramers R2                  | 0.8412          |
| [T3] hot/cold amp                | 2.0093          |
| [T3] dU barrier                  | 5.7106          |
| [T4] T_star_B                    | -0.0009         |
| [T4] proxy? (1=yes)              | 0.0000          |
| [T4] VVIX_star                   | 142.9400        |
| [T5] corr E_hi resid             | 0.2017          |
| [T6] unify raw (trading)         | 0.4963          |
| [T6] unify ema (trading)         | 0.6718          |
| [T6] unify daily                 | 0.6751          |
| [T6] unify cross                 | 0.6314          |
| [T6] delta unify                 | 0.0404          |
| [T6b] Kramers prob hi            | 0.1334          |
| [T6b] Kramers prob lo            | 0.1082          |
| [T6b] spearman P vs y            | 0.0358          |
| [L6] health std                  | 0.2887          |

```
===== DUAL-TEMP CERT v3.3 FINAL (Term-Structure T curve) =====
Trading cov: 38.4% | T_curve gate: 91% R2med: 0.82
b_med=0.0009 backwardation%=8
-----
[C1] EQ-alignment corr_lf=0.829 (thr 0.3) -> PASS-EQ-ALIGNED
[C2] Thermo=Field ema=0.672 delta=0.040 -> PASS
[C3] NEI rho=0.118 adf_p=0.000 -> PASS
-----
VERDICT: *** DUAL-TEMPERATURE FRAMEWORK CONFIRMED ***
NEI: hi=103967min fwdvol top/bot=0.4544/0.3193
T*= -0.0009 VVIX*=143 KramR2=0.841
=====
```

**Figure 2. The live dual-temperature pipeline results**



**Figure 3.** The live dual-temperature pipeline results visualization

### 3.2 Methods

All per-window estimation follows Paper III's pipeline. NEI residualization follows Sun (2026e): orthogonalization against GARCH conditional volatility and Amihud illiquidity, so that "NEI" is interpretable as the thermodynamic component of noise exposure. Newey–West (1987) standard errors with automatic lag selection throughout. No result in this paper uses data after 2026-09.

## 4. Verdicts

### 4.1 P1 (mass softening): REFUTED

Paper IV required  $\gamma < 0$  and monotone scatter. With 311 windows and the live minute test, the relation is estimated as  $m^2 = 0.0010 + 0.0000 \cdot T$  with  $t = -0.0$  — a coefficient indistinguishable from zero at roughly forty times Paper IV's effective sample size, with the scatter remaining non-monotone. **Verdict: the Debye-screening analogy fails in this system.** Mean-reversion strength of the VIX microstructure is not linearly eroded by VVIX. What would resurrect P1: a non-linear (threshold or regime-dependent) specification with economic justification, or an alternative mass estimator (e.g., jump-robust curvature) — neither currently motivated by the scatter.

### 4.2 P2 (thermal Kramers law): SUPPORTED IN SIGN

The log-crossing-rate/ $\Delta U$ -T relation is positive in **100% of 311 windows** at daily aggregation ( $R^2 = 0.58$ – $0.63$  by year: 0.582, 0.584, 0.634, 0.586) and at  **$R^2 = 0.831$**  in minute event-time in the live pipeline (M3). The direction is thus the most robust fact in the arc. Two qualifications are material:

1. **Aggregation dependence.** The measured  $R^2$  is a joint property of the law and the estimator: daily window OOS  $\approx 0.6$ ; minute event-time  $\approx 0.84$ ; Paper IV's 7-bin windows 0.55–0.90 were sampling noise around this aggregation-dependent pair.
2. **Amplification revision.** Paper IV reported extreme-hot/extreme-cold crossing-rate contrasts "of one to two orders of magnitude." The live pipeline measures hot/cold amplification = **2.01** $\times$ . We regard the live figure — computed with estimators on continuous minute data — as the better estimate and withdraw the orders-of-magnitude claim. The qualitative content (escape is thermally accelerated) stands; the quantitative drama does not.

### 4.3 P3 (critical temperature): NON-IDENTIFIABLE AS POSED

Three estimates now exist: VVIX  $\approx 90$ – $110$  (Paper IV, historical windows), VVIX = **131** ~ **143** (live pipeline), and a z-scale range of  $-1.22$  to  $+1.73$  across Paper IV's windows. We argue the instability is not a power problem but an **identification** problem:

**Proposition ( $T^*$  non-identification under rolling normalization).** Let  $T(t) =$  z-score of  $X(t)$  over trailing window  $W$ . If  $E[X]$  is non-stationary over the horizon of interest — here, VVIX has ADF  $p = 0.000$  but a half-life of 103,967 minutes ( $\approx 9$  trading months at 38.4% coverage) — then any level  $x$  of  $X$  mapped to a constant critical z-score  $z$  implies  $z(t) = (x - \mu_W(t))/\sigma_W(t)$ , which drifts with  $\mu_W$ . A "critical temperature" estimated in z-units is therefore a property of the normalization window, not of the system.

Consistently, the live pipeline estimates  $T_B^* = -0.0009$ : at the putative critical point the barrier itself is flat ( $\Delta U = 5.7106$  with near-zero temperature sensitivity). The double well

does not degrade at a point; the estimator of the well degrades over a region. **Verdict:  $T$  is not located and cannot be located within rolling z-score temperature.**

#### 4.4 P4 (RG-flow reversal): *REFUTED*

Paper IV observed higher-mode energy share declining with temperature in most episodes windows and offered two interpretations: (a) genuine microstructure — stress concentrates dynamics in the level mode as dealers widen quotes and granular price discovery collapses; (b) artifact — hot bins contain fewer, more autocorrelated observations. The live pipeline, with different aggregation and different period, measures RG residual correlation = **0.202**, which shown to have regime depend and drifting properties. The companion diagnostic paper’s panic result —  $T_s$  declining monotonically from  $-0.003$  (2023) to  $-0.128$  (2026), the parallel-shift signature — supplies exactly the potential-landscape geometry that interpretation (a) requires: in stress, the landscape shifts in parallel and dynamics collapse onto the level coordinate.

**Verdict: P4 is refuted as a criticality claim.** The VIX microstructure field does not approach a Wilsonian critical point under VVIX heating; it approaches a **single-mode absorbing state**.

#### 4.5 Unification: *PARTIAL, WITH A STRUCTURAL CEILING*

The correlation between field-derived Gibbs free energy and directly computed  $G$  rises from Paper IV’s mean  $0.024$  to **0.6751** (daily; EMA variant  $0.6718$ ; raw trading  $0.4936$ ; cross  $0.6314$ ; delta-unify  $0.0404$ ) under the dual-temperature pipeline. We grade this against the ceiling conditional on the regime specified and set by thermometer agreement,  $\text{corr}(T_C, T_K) = 0.8288$ : the observed unification ( $R^2 \approx 0.6751$ ) sits well below the ceiling ( $R^2 \approx 0.8288$ ), and the residual correlates with the illiquidity-residual NEI of Sun (2026e) ( $\text{NEI}_r$  ranging  $-0.003$  to  $-0.144$ , sign-flipping by regime). **Verdict: Paper IV’s strong claim — “ $G$  is derived, not posited” — is withdrawn. The weak claim — thermodynamics and field statistics share substantial common variance, with a measurable, economically interpretable gap — is affirmed.** The gap is itself informative: it is the portion of market “heat” that is neither implied-vol nor term-structure-implied, i.e., the liquidity-fragmentation component.

#### 4.6 The tradeable signal:

The top-quintile realized crossing rate **0.1334** vs. bottom-quintile **0.1082**; rank correlation  $\text{spearman}(P, y) = \mathbf{0.0358}$  — a precision null.

The control comparison is the finding. Sorting the identical forward-vs-realized volatility spread on NEI yields  $\text{hi/lo} = \mathbf{0.4544 / 0.3193}$  (spread  $+0.1351$ ; NW  $t = 6.35$  in the 311-window archive), while sorting on the **Kramers transition probability** yields  $0.03258$  with  $\rho \approx 0.036$ . Both objects are functions of the same thermal primitives; only the state variable prices. **Verdict: the one-day Kramers transition probability, as constructed, carries no predictive content. We conjecture — and pre-specify for future work — that transition probabilities over this horizon are dominated by estimator noise in  $\Delta U(t)$ , whereas the NEI state variable aggregates the same information with lower variance.**

## 5. Discussion

### 5.1 The falsification program worked

Every prediction in Paper IV was specified with a functional form and a kill condition; every one has now been killed, qualified, or re-specified by evidence gathered under stricter discipline than the original. This is the arc functioning as designed, and it is the strongest available answer to the econophysics credibility critique: not that the analogies were right, but that they were built to be shown wrong.

### 5.2 *What survives*

Three objects survive the verdicts with enhanced standing: (i) the **sign** of the thermal Kramers relation, 100% stable across 311 windows and an independent live pipeline; (ii) the **NEI state variable**, with OOS spread economics at NW  $t = 6.35$  and live replication (spearman 0.181, inside the historical 0.089–0.288 band); (iii) the **dual-temperature fact**  $\text{corr}(T_C, T_K) = 0.8288$ , which Paper IV anticipated qualitatively and which now anchors the revised theory of the companion paper (Sun, 2026f).

### 5.3 *Limitations*

The unification residual's NEI correlation is suggestive, not yet a formally identified decomposition.

## 6. Conclusion

We executed the validation program that Paper IV prescribed for itself, with forty times its effective sample size and an independent live pipeline. The verdicts: mass softening is refuted; the Kramers law survives in sign; the critical temperature is non-identifiable under rolling normalization; the RG-flow reversal is refuted; unification is partial and structurally capped; the tradeable signal is a precision null while its state-variable sibling prices at  $t = 6.35$ . The temperature program, honestly audited, has yielded one robust law, one priced state variable, and a partial unification claim. That is a normal, useful scientific harvest — and a better one than the theory deserved and received in its untested form.

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30 **Competing interests:** †Core technical implementation protected by Chinese Patent Application No. CN202511140743.3A (Filed: 2025-08-14) and Patent Cooperation Treaty (PCT) international patent application No. PCT/CN2026/112498 (Filed: 2026-07-22). Full patent documentation available via CNIPA/WIPO.

35 **Data, code, and materials availability:** All data, code, and materials used in the analysis are available upon requests and signing the materials transfer agreements (MTAs) for purposes of reproducing or extending the analysis.